

Module 15: Population and Systems Ecology — Questions

Introduction

The following questions cover the fundamental interactions between organisms and their environments, from mathematical growth models to the movement of energy across global ecosystems. Use these to test your understanding of how biological systems operate and sustain themselves.

Practice Questions

1. What does a population graph experiencing exponential growth look like, and under what conditions does it occur?
2. Why is exponential growth rarely sustained over long periods of time in the real world?
3. Describe the shape of a logistic growth curve and explain what causes the curve to flatten out.
4. Define the ecological concept of carrying capacity (often designated as K).
5. Give three examples of density-dependent factors that can limit population growth as the population size approaches the carrying capacity.
6. What is a density-independent factor, and how does it affect populations differently than density-dependent factors?
7. Explain the role of primary producers at the base of an ecosystem's food web.
8. Distinguish between a primary consumer, secondary consumer, and apex predator.
9. What crucial ecological role do decomposers and detritivores play in nutrient cycling?
10. State the 10% Rule of Energy Transfer in ecosystems.
11. According to the 10% rule, why are there fundamentally fewer apex predators in a given area compared to plants or herbivores?
12. Where does the "lost" 90% of energy primarily go as you move up each trophic level?
13. How does the flow of energy through an ecosystem fundamentally differ from the cycling of vital chemical nutrients (like carbon, nitrogen, and phosphorus)?

14. Briefly describe the role of photosynthesis and cellular respiration in the global Carbon Cycle.
15. Why do farmers use fertilizer, knowing what we know about how elements like nitrogen cycle locally?